Eric Martin Feltham

eric.feltham@aya.yale.edu | 0000-0001-8080-7119 | emfeltham | emfeltham.github.io

Last updated: August 2026

Employment

Data Science Institute Postdoctoral Fellow — **Sept. 2024–Present**
Columbia University — *New York, NY*

Postdoctoral Associate — **June 2023–Aug. 2024**
Yale Institute for Network Science and Department of Sociology, Yale University — *New Haven, CT*

Education

Ph.D., Sociology (with Distinction), Yale University — **May 2023**
Dissertation: "Cognizing Social Networks" — *New Haven, CT*
Committee: Drs. Nicholas Christakis (chair), Emily Erikson, Laura Forastiere, Ifat Levy
Winner of the 2024 Marvin B. Sussman Dissertation Prize from Yale University

M.A., Statistics, Yale University — **2018**
New Haven, CT

B.A., Philosophy; B.A., Economics, University of Massachusetts Amherst — **2013**
magna cum laude, *Phi Beta Kappa* — *Amherst, MA*

Research interests

Social network cognition: how people map social structure, who they believe is tied to whom, and what these representations mean for influence, diffusion, and collective action in groups and organizations. Related interests: social theory; intergroup boundaries, coordination, and conflict. Methods: computational social science, causal inference, and experiments.

Publications

Peer-reviewed articles

Eric Feltham, Laura Forastiere, and Nicholas Christakis. "Cognitive representations of social networks in isolated villages". *Nature Human Behaviour* 9, 1737–1753 (2025). https://doi.org/10.1038/s41562-025-02221-6

- Selected for a "Research Briefing" feature by *Nature Human Behaviour*. https://doi.org/10.1038/s41562-025-02222-5
- Featured in Phys.org ("Biases shape how people mentally represent social ties in their community, study suggests", June 2025)

Eric Feltham, Laura Forastiere, Marcus Alexander, Nicholas Christakis. "Mass gatherings for political expression had no discernible association with the local course of the COVID-19 pandemic in the USA in 2020 and 2021". *Nature Human Behaviour* 7, 1708–1728 (2023). https://doi.org/10.1038/s41562-023-01654-1

- Featured in *FiveThirtyEight*.

Philipp Homan, Ifat Levy, **Eric Feltham**, Charles Gordon, Jingchu Hu, Jian Li, Robert H. Pietrzak, Steven Southwick, John H. Krystal, Ilan Harpaz-Rotem, Daniela Schiller. "Neural computations of threat in the aftermath of combat trauma". *Nature Neuroscience* 22, 470–476 (2019). https://doi.org/10.1038/s41593-018-0315-x

Book chapters

Emily Erikson and **Eric Feltham**. "Structure". In *Research Handbook on Analytical Sociology*. Cheltenham, UK: Edward Elgar Publishing (2021).

Emily Erikson and **Eric Feltham**. "Historical Network Research". In *Oxford Handbook of Social Networks*. New York: Oxford UP (2020). https://doi.org/10.1093/oxfordhb/9780190251765.001.0001

Working papers

Unpublished drafts are available at https://emfeltham.github.io/working-papers/

Eric Feltham and Nicholas Christakis. "Cognitive Expectations of Homophily in Village Social Networks". Under review. SocArXiv preprint (2026). https://osf.io/preprints/socarxiv/z4nyq_v3

Eric Feltham. "Network Cognition in Cultural Context".

Eric Feltham. "Signal Degradation without Convergence: Scale Relevance and the Politics of a Credential".

Eric Feltham. "The Decline in White Americans' Implicit Racial Bias Diverged Along the Geography of Manufacturing Decline".

Eric Feltham and Nicholas Christakis. "Seeing Structure Similarly".

Eric Feltham. "The Yes Rate Revisited: PPB as a Distribution-Free Complement to Youden's J". Under review.

Eric Feltham. "FormulaCompiler.jl and Margins.jl: Efficient Marginal Effects in Julia". Under review. arXiv preprint (2026). https://arxiv.org/abs/2601.07065

Eric Feltham. "empirica-networks: online network experiments with Empirica".

Research in progress

Eric Feltham, Laura Forastiere, Johan Ugander, and Nicholas Christakis. "Off-Policy Evaluation with Cognitive Social Structures".

Eric Feltham and Nicholas Christakis. "The Networked Climate Game".

Eric Feltham. "Off the Party Menu: Beliefs and Within-Party Hostility, 1972–2024".

Dominik Duell, **Eric Feltham**, Nicholas Christakis. "Revolutions *in silico*".

Eric Feltham. "TSCSMethods.jl: Extended Matching for Generalized Difference-in-Differences with Time-Series Cross-Sectional Data".

Book project

"Cognizing Social Networks" (working title). Manuscript in preparation; proposal under review.

Grants

Research Grant (R01), "Characterizing Individuals' Cognitive Maps of their Village Social Networks" **2022**

National Institutes of Health, The National Institute on Aging and the Office of Behavioral and Social Sciences Research

- Lead author of the proposal, research design, and analysis plan. Award: $3,226,809 over 4 years. PI: Dr. Nicholas Christakis.

"Assessing the Hazard of Elections During the COVID-19 Pandemic" **2020**

Robert Wood Johnson Foundation

- Lead author of the proposal, research design, and analysis plan. Award: $350,000. PI: Dr. Nicholas Christakis.

Honors and awards

Sussman Prize for best dissertation in the past two years **2024**
Yale Department of Sociology, New Haven, CT

Commonwealth Honors College Scholar with Greatest Distinction **2013**
University of Massachusetts Amherst, Amherst, MA

Invited talks

"Cognizing Social Networks"

- **Santa Fe Institute**, April 2026. Recording: https://emfeltham.github.io/talks/
- **Stanford Graduate School of Business**, Organizational Behavior Area, November 2025
- **Harvard University**, Department of Psychology (Cognition, Brain, Behavior Seminar), September 2025
- **Yale University**, Department of Sociology (Sussman Award Lecture), May 2025

Other talks

- **Yale University**, Human Nature Lab — "Expecting Homophily", January 2026
- **University of Iowa**, Department of Sociology (Social Networks Workshop) — "Cognizing Social Networks", November 2023
- **Yale University**, Center for Empirical Research on Stratification and Inequality — "Modeling the Effect of Identity on Polarization", January 2019
- **Yale Institute for Network Science** — "Experiments on Observational Learning", August 2018

Conference presentations

"Cognizing Social Networks"

- International Conference on Computational Social Science, Norrköping, Sweden, July 2025
- Annual Meeting of the American Sociological Association (Sections on Social Psychology and Mathematical Sociology), Montréal, QC, August 2024
- Duality@50: Making progress and looking forward, ETH Zürich, Ascona, Switzerland, April 2024
- Implementation and Dissemination Science Workshop, Copán, Honduras, June 2023

"Expecting Homophily"

- Annual Meeting of the American Sociological Association (Sections on Social Psychology and Mathematical Sociology), New York City, August 2026
- International Conference on Computational Social Science, Burlington, VT, July 2026
- Annual Conference of the International Network for Social Network Analysis (Sunbelt 2026), Daytona Beach, FL, June 2026
- International Conference on Network Science (NetSci 2026), Boston, MA, June 2026
- Annual Conference of the International Network of Analytical Sociology, Columbia University, June 2025

Megha Chawla, Philipp Homan, Charles Gordon, **Eric Feltham**, Daniela Schiller, Ilan Harpaz-Rotem, Ifat Levy. **"Cortical thickness and volume of the right posterior parietal cortex predict individual learning rate in healthy adults"**

- Society for Neuroscience, San Diego, CA, 2018

Software

Margins.jl and FormulaCompiler.jl (sole author)

- Computes marginal effects for generalized linear models and mixed models
- Provides a unified, typed compilation framework for statistical model formulas in Julia

TSCSMethods.jl (sole author)

- Performs nonparametric generalized difference-in-differences estimation, with covariate matching

SamplingPerceivedNetworks.jl (sole author)

- Implements a sampling procedure for "cognitive social structures" data collection

Contributions to open-source packages: NetworkLayout.jl and GraphDataFrameBridge.jl.

Other writing

"Social network cognition among isolated villagers reveals distinct patterns of bias". *Nature Human Behaviour Research Briefing* (June 2025). (Invited contribution by Nature Human Behaviour)

Eric Feltham and Nicholas Christakis. "Rapid research with a pandemic bearing down: Studying the impact of mass gatherings on the course of COVID-19". *Springer Nature Communities* (2023). (Invited contribution to "Behind the Paper" series for the Springer Nature Social Sciences online community)

Eric Feltham and Nicholas Christakis. "Voting In The 2020 Primaries Didn't Worsen The COVID-19 Pandemic". *FiveThirtyEight* (2020).

Teaching

Teaching Fellow, Yale College, Yale University (2018–2023):

- SOCY 133a, Computers, Networks and Society (Fall 2018)
- SOCY 167b, Social Networks and Society (Spring 2019)
- SOCY 151a, Foundations of Modern Social Theory (Fall 2019)
- SOCY 126b, Health of the Public (Spring 2020)
- SOCY 134a, Sex and Gender in Society (Fall 2021)
- SOCY 152b, Topics in Contemporary Social Theory (Spring 2022)
- SOCY 126a, Health of the Public (Fall 2022)
- SOCY 141b, Sociology of Crime and Deviance (Spring 2023)

Mentoring

- Ashley Oaks (Statistics & Data Science, Yale University, PhD)
- Shriya Muthukumar (Computer Science & Economics, University of Chicago, 2029)
- Aslı Yüksel (Psychology, Swarthmore College, 2026)
- Karen Gao (Computer Science & Psychology, Yale College, 2027)
- Breanna Nguyen (Cognitive Science, Yale College, 2025)
- Kaity Chandrika (Sociology, Yale College, 2025)
- Justice Harasha (Economics & Mathematics, Yale College, 2024)
- Kayla Morgan (Cognitive Science, Yale College, 2024)
- Benjamin Snyder (Economics, Yale College, 2022)

Research experience

Graduate Researcher **2016–2023**

Yale Institute for Network Science and Human Nature Lab, New Haven, CT

- Designed and led large-scale data collection and analysis efforts on individuals' perceptions of social network structure with over 10,000 people in 82 villages in rural Honduras
- Developed sampling method for cognitive social structures research design
- Conducted fieldwork in rural Honduras to support data collection and analysis effort for thesis project
- Developed generalized difference-in-differences methodology to estimate the impact of large-scale political gatherings on the spread of COVID-19 in the USA

Research Fellow **2019**

Dr. Emily Erikson, Yale University, New Haven, CT

- Analyzed historical social network data consisting of English economic manuscript writers from the 18th century
- Applied topic modeling to uncover trends in the economic literature over a roughly 200-year period

Graduate Researcher **2018**

University of Chicago, Chicago, IL

- Part of a team on the grant "Social MIND: Social Machine Intelligence for Novel Discovery", with Drs. James Evans and Michael Franklin of the University of Chicago and Dr. Joshua Tenenbaum of MIT

Research Assistant (fMRI and fear conditioning) **2014–2016**

Levy and Harpaz-Rotem Labs, Yale School of Medicine, New Haven, CT

Research Assistant (human fear conditioning) **2014**

Schiller Affective Neuroscience Lab, Icahn School of Medicine at Mount Sinai, New York, NY

Professional service

Memberships. American Sociological Association; International Network for Social Network Analysis
Ad hoc reviews. Proceedings of the National Academy of Sciences of the United States of America; PNAS Nexus; Sociological Science; Risk Analysis; Policy Analysis; Group Processes & Intergroup Relations; Rationality and Society; Yale Undergraduate Research Journal

Skills

Programming. Julia, R, MATLAB, Python. **Languages.** Spanish (basic), French (basic)

Leadership & service

Campaign Coordinator **2010–2011**

Students for a Just and Stable Future, Boston, MA

- Built a statewide student climate coalition; contributed to MA climate policy

Team Leader **2010–2011**

Better Future Project, Boston, MA

References

1. Dr. Peter Bearman, Jonathan R. Cole Professor of the Social Sciences, Columbia University (psb17@columbia.edu)
2. Dr. Nicholas Christakis, Sterling Professor of Social and Natural Science, Yale University (nicholas.christakis@yale.edu)
3. Dr. Emily Erikson, Professor of Sociology and (by courtesy) Management, Yale University (emily.erikson@yale.edu)
4. Dr. Laura Forastiere, Associate Professor of Biostatistics, Yale School of Public Health, Yale University (laura.forastiere@yale.edu)
5. Dr. Ifat Levy, Elizabeth Mears and House Jameson Professor of Comparative Medicine, with secondary appointments in Psychology and in Neuroscience, Yale School of Medicine, Yale University (ifat.levy@yale.edu)